Policy-BT: from workspace evidence to evolutionary feedback

One task, one fixed comparison contract; a growing tournament of generation workspaces.

A Admit and normalize

Generation workspace

Scripts · logs · tables
Goal-relevant figures

Snapshot + reproduction

Required shell-script contract
Fresh-venv rerun by default

Ineligible → reward 0

pass

Reconnaissance + adapters

Goal → fixed, ordered criteria
Artifact schemas → shared values
Measured values; unknown = null

B Decide each workspace pair

Workspace A values

Workspace B values

Cached Python comparison policy

Same task criteria for both workspaces
Earliest decisive stage determines the verdict
Reverse-order, self-tie + determinism checks

Only if code ties

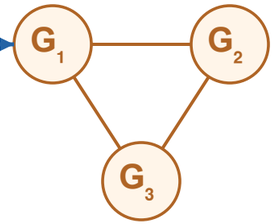
Image-based tie resolution

Requires relevant images on both sides
Judge A/B and B/A; winners must agree
Absent, unclear or inconsistent → keep tie

Code decides

A wins / B wins / tie + evidence

C Rate and guide evolution



All pairs
in the task
session

Regularized Bradley–Terry fit

Pair outcomes → latent strengths θ
$P(i \text{ beats } j) = \sigma(\theta_i - \theta_j)$
Current reward: $r_i = \sigma(\theta_i)$ if eligible

Refit every generation's current reward after each new evaluation.

Evidence-grounded feedback

Earliest losses first; no scores or code

Selection + mutation → next generation

Evolutionary pressure, not a benchmark grade. σ = logistic function. Admission verdicts also enter the tournament.